

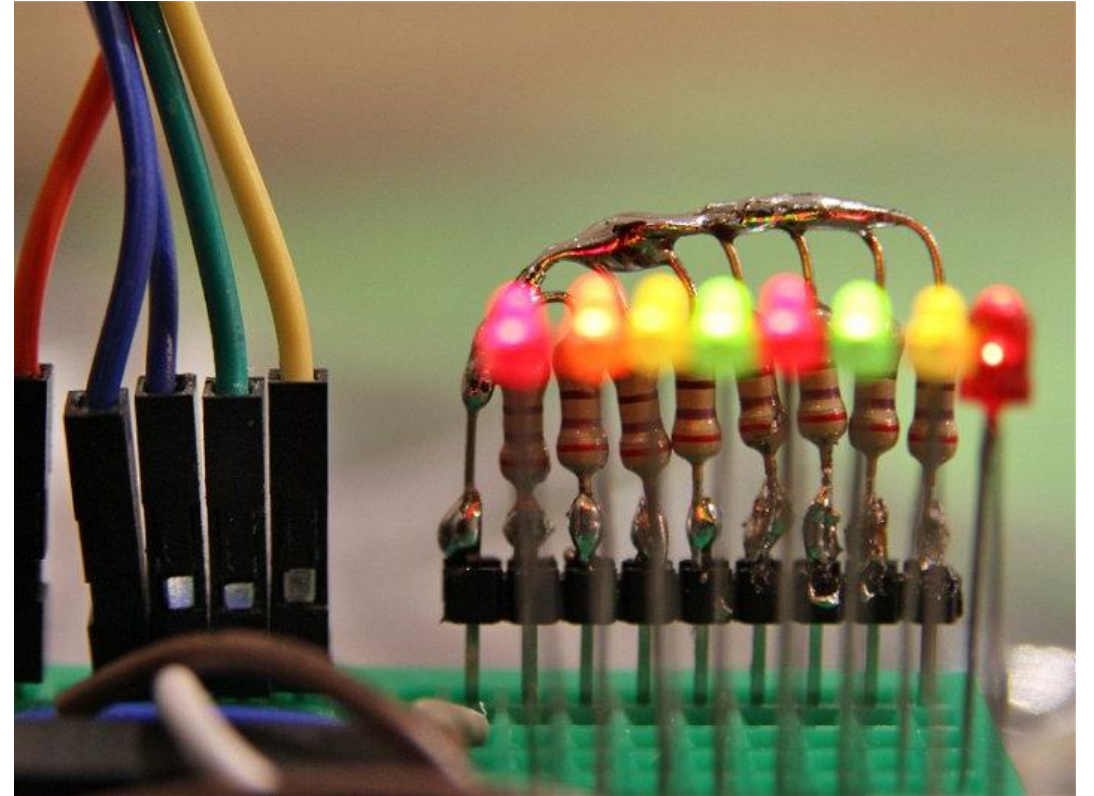
Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 3

Resistive Network Analysis

Principles and Applications of Electrical Engineering, 7th ed., Chapter 2
Electric Circuits, 11th ed., Chapter 3



2.1 Resistors in Series and Voltage Division

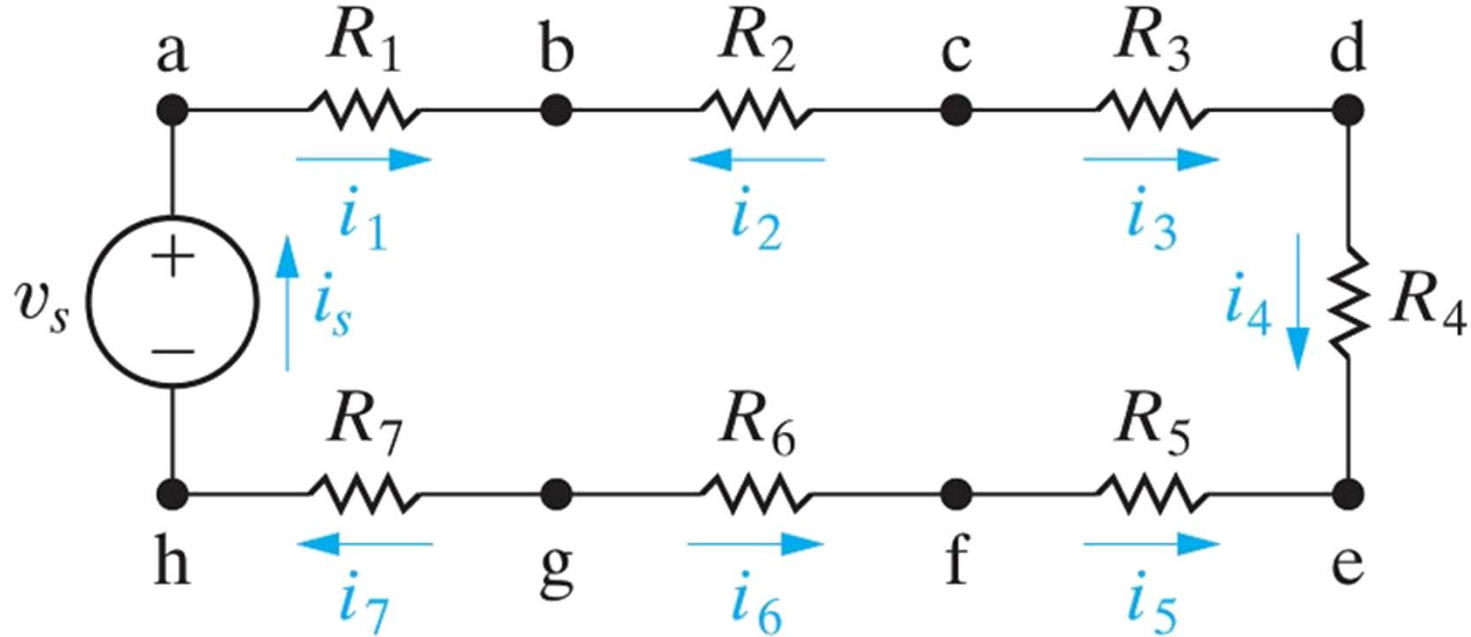
Principle of voltage division.

- The ratio of the voltages across any two series resistances equals the ratio of those resistances.
- Applies to any two resistances in series.

Equivalent resistance equals sum of all resistances for resistors in series.



2.1 Resistors in Series and Voltage Division



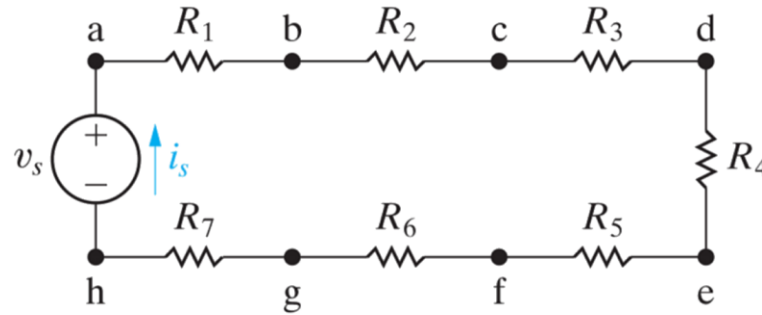
Resistors in series:

- Connected end-to-end
- Have the same current (because of KCL)

2.1 Resistors in Series and Voltage Division

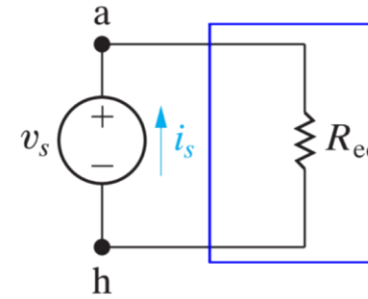
Resistors in series:

- Find the equivalent resistance



$$\text{KVL} : -v_s + R_1 i_s + R_2 i_s + \cdots + R_7 i_s = 0$$

$$\Rightarrow v_s = (R_1 + R_2 + \cdots + R_7) i_s$$



$$\text{KVL} : -v_s + R_{eq} i_s = 0$$

$$\Rightarrow v_s = (R_{eq}) i_s$$

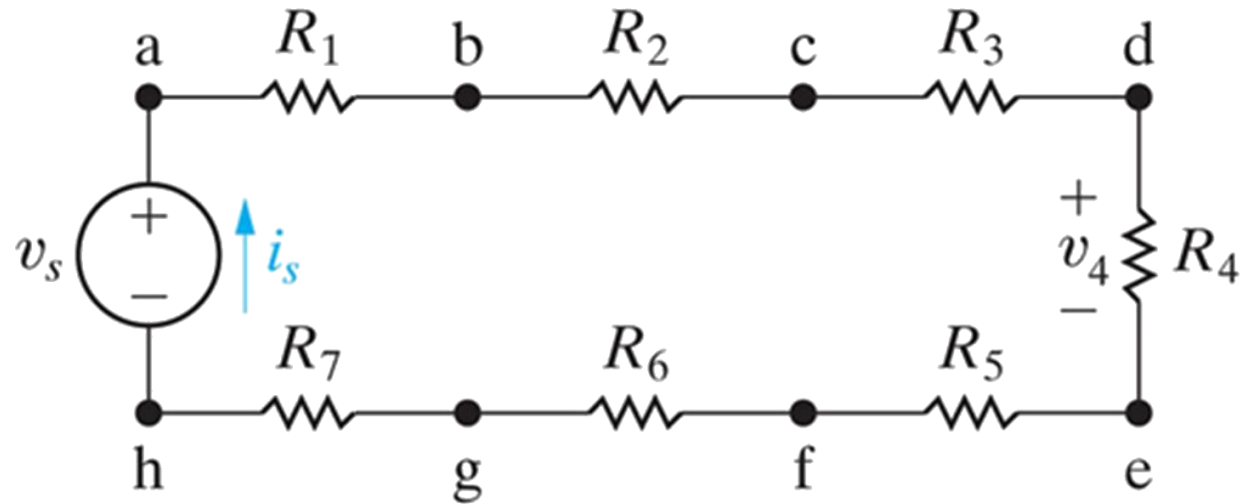
Resistors in series add: $R_{eq} = (R_1 + R_2 + \cdots + R_7)$

- Equivalent resistance is always larger than the largest resistance in the group.

2.1 Resistors in Series and Voltage Division

Voltage division:

- Find the voltage drop across one of a group of series-connected resistors



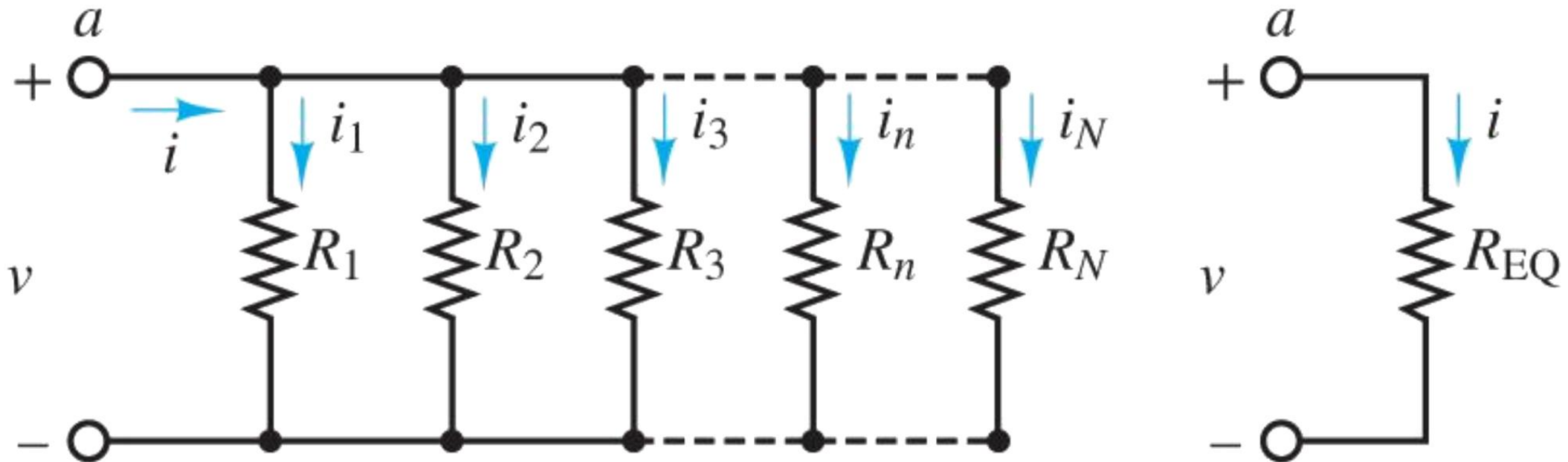
$$v_4 = R_4 i_s = R_4 \frac{v_s}{(R_1 + R_2 + \cdots + R_7)} = \frac{R_4}{R_{eq}} v_s$$

2.2 Resistors in Parallel and Current Division

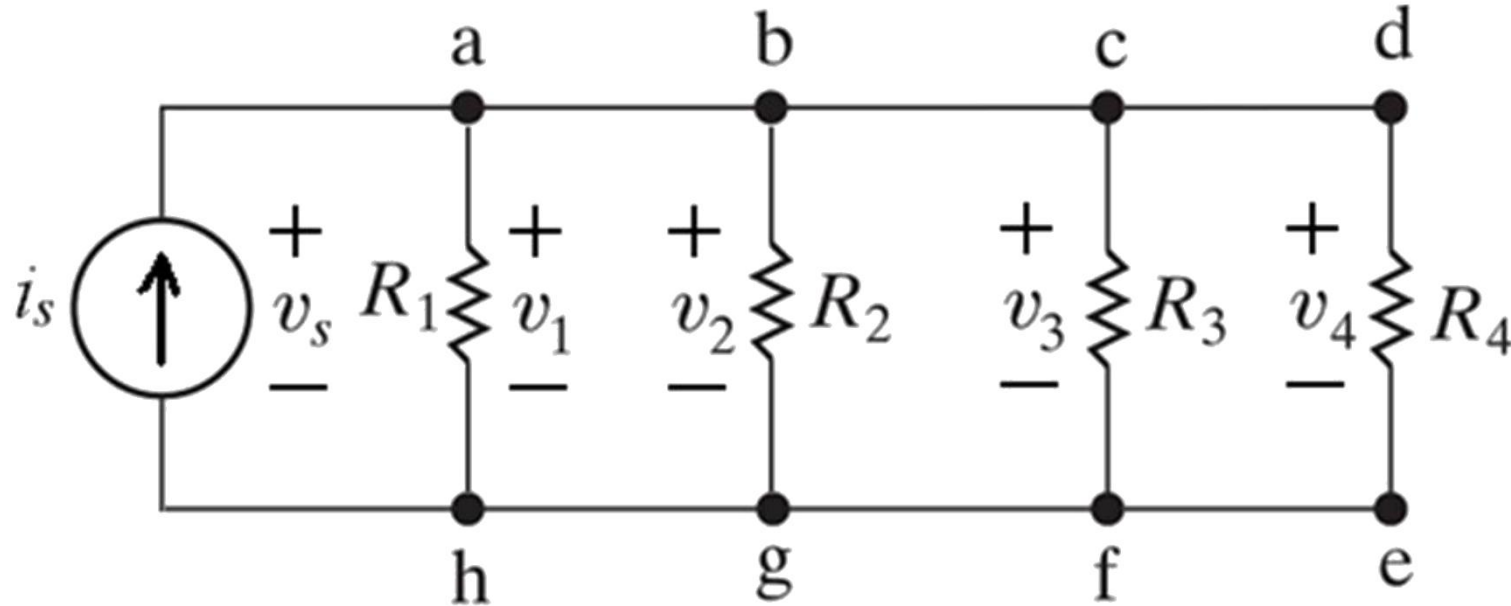
Principle of current division.

- The ratio of the currents through any two resistances in parallel equals the inverse ratio of those resistances.

Equivalent resistance of parallel resistors.



2.2 Resistors in Parallel and Current Division



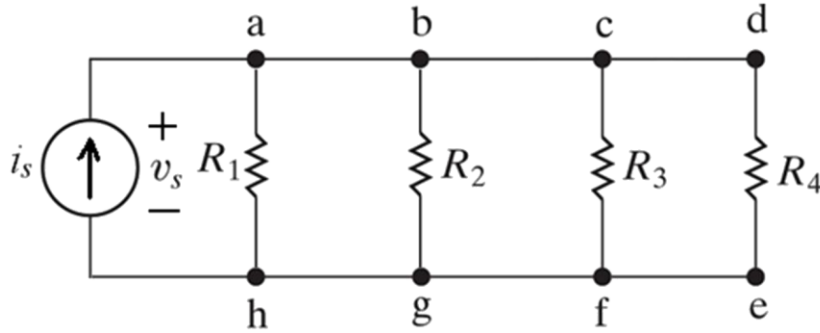
Resistors in parallel:

- Connected at both ends
- Have the same voltage (because of KVL)

2.2 Resistors in Parallel and Current Division

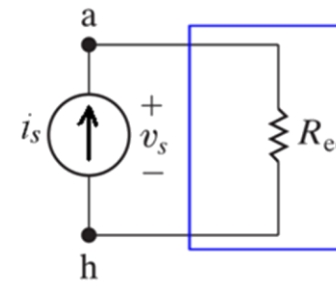
Resistors in parallel:

- Find the equivalent resistance



$$\text{KCL: } -i_s + v_s/R_1 + v_s/R_2 + \cdots + v_s/R_4 = 0$$

$$\Rightarrow i_s = (1/R_1 + 1/R_2 + \cdots + 1/R_4)v_s$$



$$\text{KCL: } -i_s + v_s/R_{eq} = 0$$

$$\Rightarrow i_s = v_s/(R_{eq})$$

Resistors in parallel:

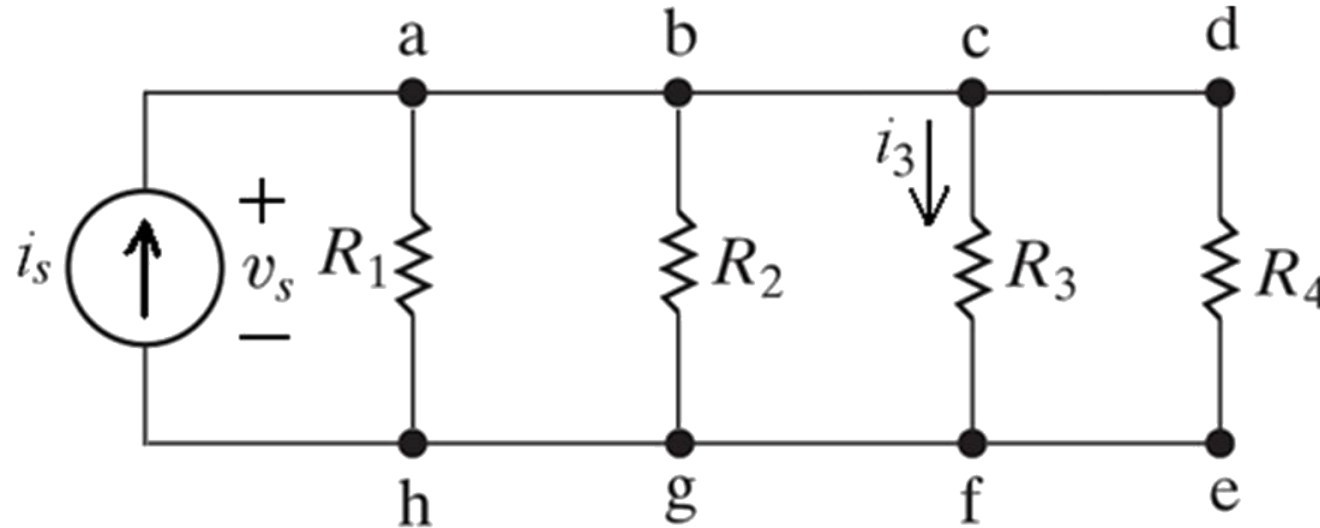
$$1/R_{eq} = (1/R_1 + 1/R_2 + \cdots + 1/R_4)$$

- Equivalent resistance is always smaller than the smallest resistance in the group.

2.2 Resistors in Parallel and Current Division

Current division:

- Find the current through one of a group of parallel-connected resistors



$$i_3 = \frac{v_s}{R_3} = \frac{R_{eq} i_s}{R_3} = \frac{R_{eq}}{R_3} i_s$$

Review: Series-Connected Components

Review:

- Series-connected components
- Resistors in series add

$$R_{eq} = \sum_{i=1}^n R_i$$

- Voltage division allows you to find the voltage across one component when you know the voltage across a collection of series-connected components:

$$v_i = \frac{R_i}{R_{eq}} v_s$$

Review: Parallel-Connected Components

Review:

- Parallel-connected components
- Add the inverse resistance of resistors in parallel to get the inverse of the equivalent resistance

$$\frac{1}{R_{eq}} = \sum_{i=1}^n \frac{1}{R_i}$$

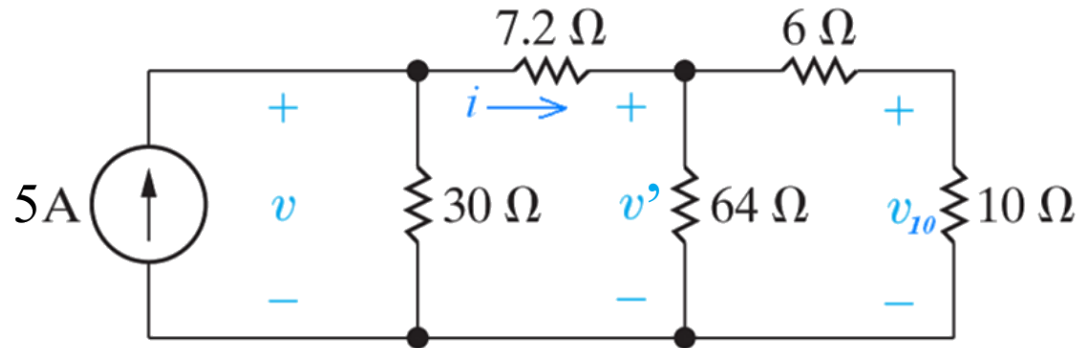
- Current division allows you to find the current through one component when you know the current supplied to a collection of parallel-connected components:

$$i_x = \frac{R_{eq}}{R_x} i_s$$

Example AP 3.1: Series-Parallel Simplification

Example AP 3.1

- Find v , i , and v_{10} by combining resistors in series and parallel, voltage division, and current division.



Find the equivalent resistance:

$$12.8 + 7.2 = 20\Omega$$

$$6 + 10 = 16\Omega$$

$$30 \parallel 20 = 12\Omega$$

$$64 \parallel 16 = 12.8\Omega$$

$$\text{Ohm's Law : } v = R_{eq}(5) = (12)(5) = 60 \text{ V}$$

$$\text{KCL : } i = \frac{R_{eq}}{R}(5) = \frac{30 \parallel 20}{20}(5) = 3 \text{ A}$$

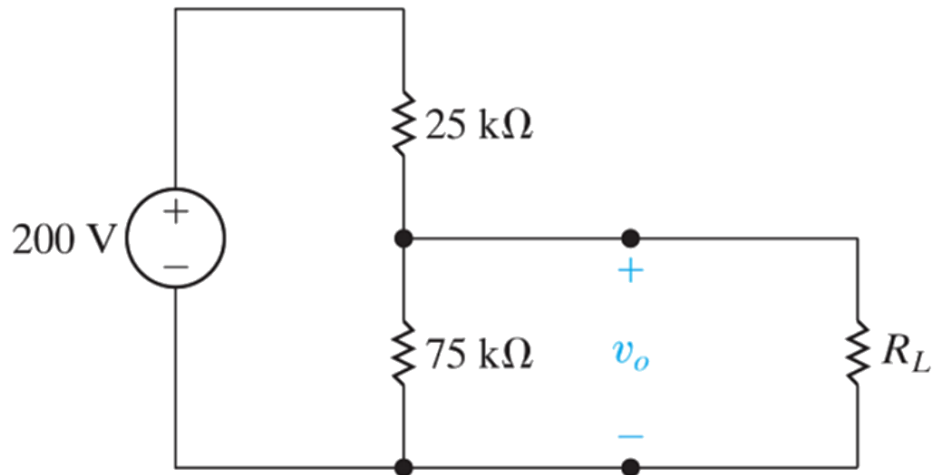
$$\text{Ohm's Law : } v' = i(64 \parallel 16) = (3)(12.8) = 38.4 \text{ V}$$

$$\text{KVL : } v_{10} = \frac{10}{10 + 6} v' = \frac{10}{16}(38.4) = 24 \text{ V}$$

Example AP 3.2: Voltage Division with a Load

Example AP 3.2

- Find v_o when there is no load resistor. Repeat for a load resistor of $150\text{ k}\Omega$.



With R_L removed, use voltage division:

$$v_o = \frac{75,000}{25,000 + 75,000} (200) = 150\text{ V}$$

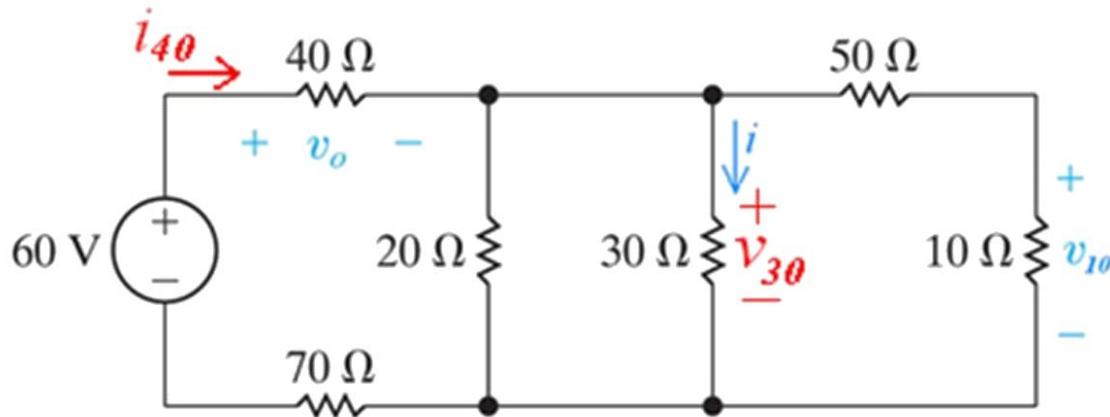
With $R_L = 150\text{ k}\Omega$, use $75\text{ k} \parallel 150\text{ k}$ and voltage division:

$$v_o = \frac{75,000 \parallel 150,000}{25,000 + 75,000 \parallel 150,000} (200) = \frac{50,000}{25,000 + 50,000} (200) = 133.33\text{ V}$$

Example AP 3.4: Series-Parallel Simplification

Example AP 3.4

- Find v_o , i , and v_{10} using series and parallel simplification, and voltage and current division.



$$20 \parallel 30 \parallel 60 = 10\Omega$$

$$50 + 10 = 60\Omega$$

$$\text{voltage division : } v_o = \frac{40}{40 + 10 + 70}(60) = 20 \text{ V}$$

$$\text{Ohm's Law : } i_{40} = \frac{v_o}{40} = \frac{20}{40} = 0.5 \text{ A}$$

$$\text{current division : } i = \frac{20 \parallel 30 \parallel 60}{30} i_{40} = \frac{10}{30}(0.5) = 0.167 \text{ A}$$

$$\text{Ohm's Law : } v_{30} = 30i = 30(0.167) = 5 \text{ V}$$

$$\text{voltage division : } v_{10} = \frac{10}{10 + 50} v_{30} = \frac{10}{60}(5) = 0.833 \text{ V}$$

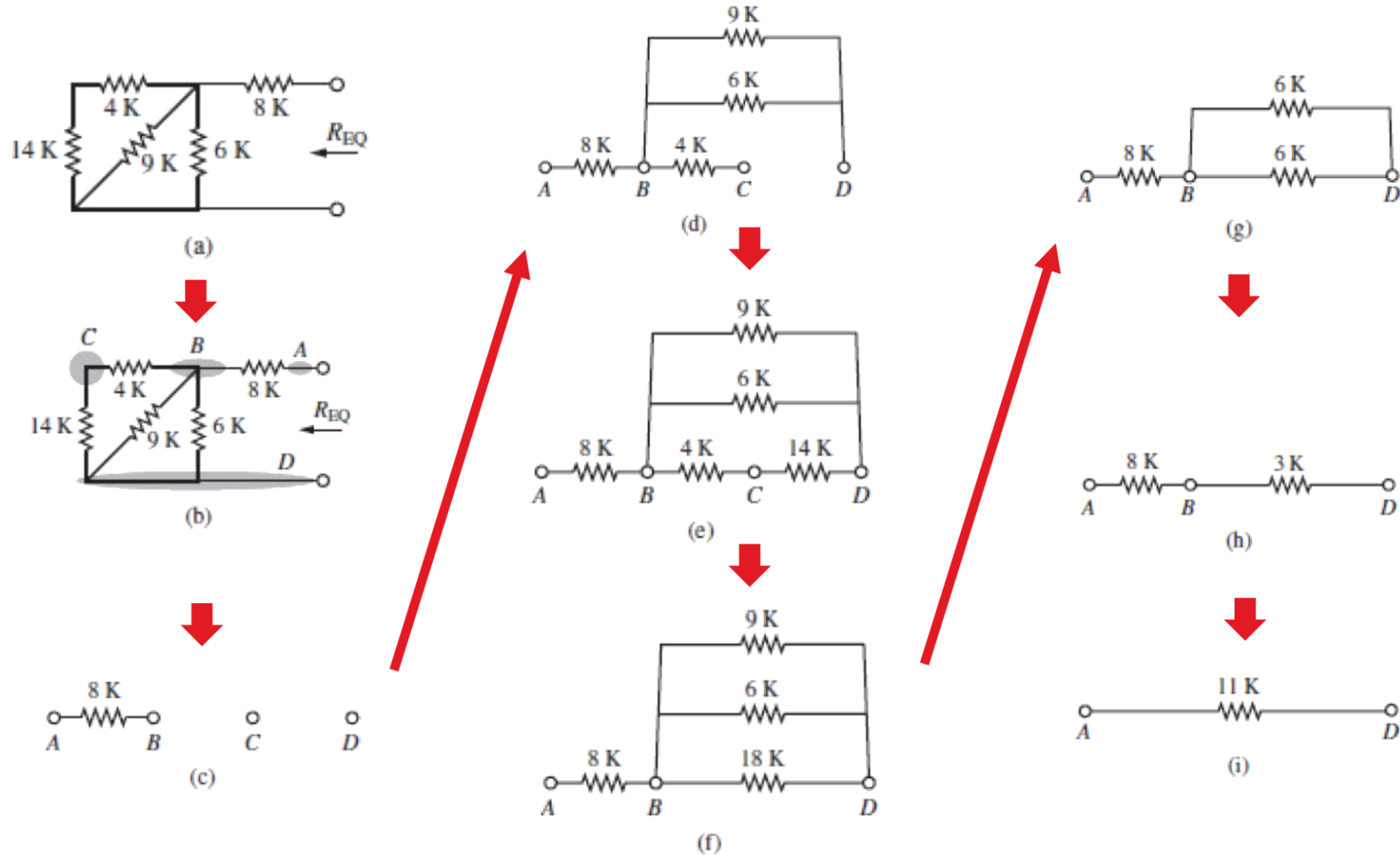
2.3 Equivalent Resistance Between Two Nodes

Two resistors in series or parallel between two nodes can be replaced by a single equivalent resistance.

- Concept used to simplify a complex network by redrawing using equivalent resistances.
- Equivalent resistance is defined by:

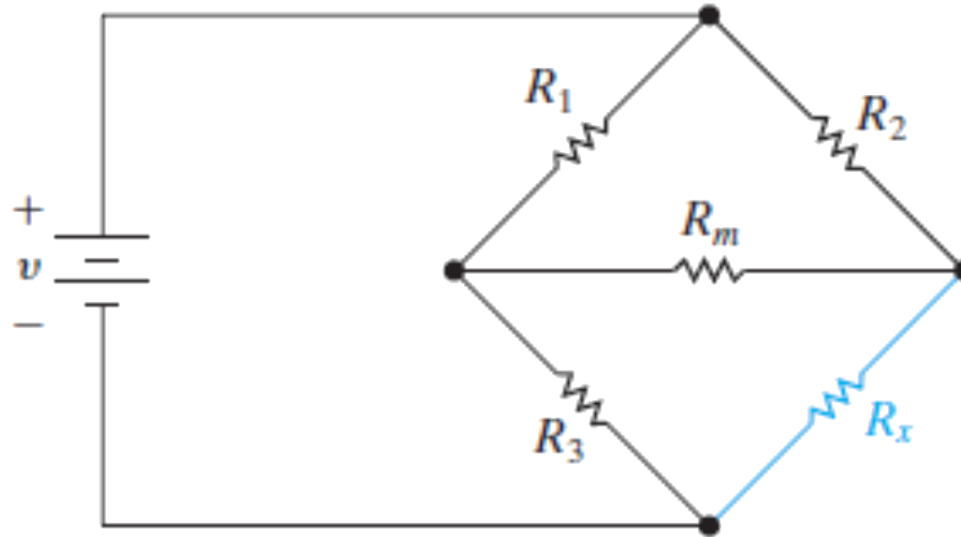
$$R_{EQ} \equiv \frac{v_s}{i_s}$$

2.3 Equivalent Resistance Between Two Nodes



2.3 Quiz: Resistors in a Bridge Circuit (1)

Which statement is true about the resistors in this bridge circuit?

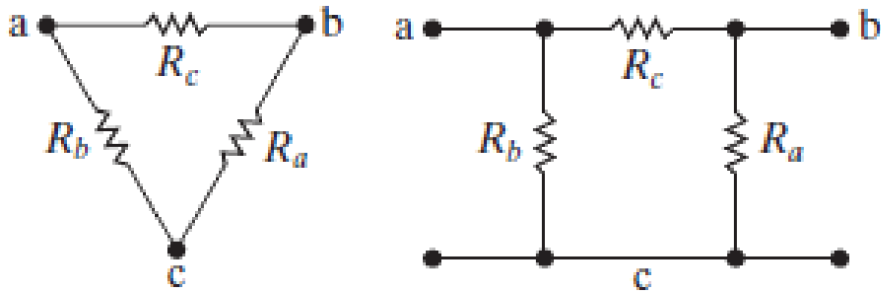


- A. R_1 and R_3 are in series
- B. R_m is in parallel with R_1 and R_2
- C. There are no parallel-connected or series-connected resistors

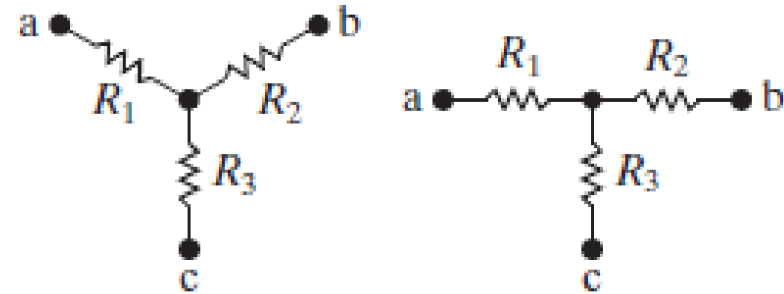
2.3 Delta-to-Wye (Pi-to-Tee) Equivalent Circuits

Delta(Δ)-to-Wye(Y) (Pi-to-Tee) Equivalent Circuits

- While the bridge circuit has no series-connected or parallel-connected resistors, it does have resistors connected in a delta (or pi) configuration and resistors connected in a wye (or tee) configuration.



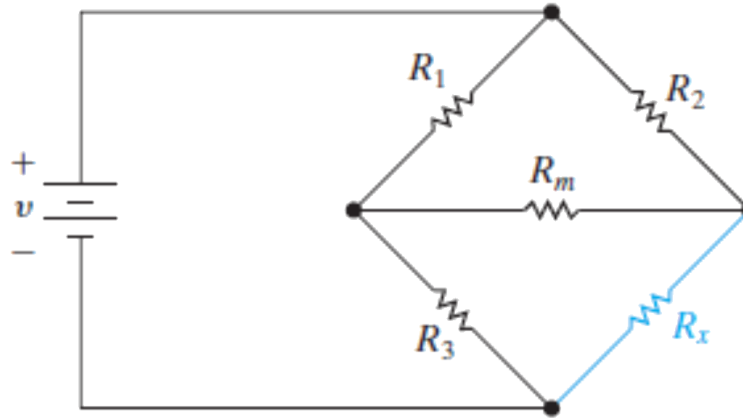
A Δ configuration rearranged to a π



A Y configuration rearranged to a T

2.3 Quiz: Resistors in a Bridge Circuit (2)

Which statement is false about the resistors in this bridge circuit?

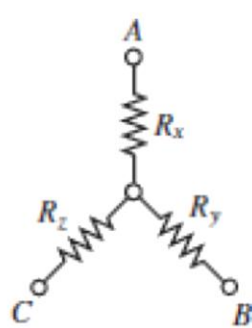


- ☒ A. R_1 , R_2 , R_m are connected in a delta
- ☒ B. R_m , R_3 and R_x are connected in a wye
- ☒ C. R_m , R_3 and R_x are connected in a delta
- ☒ D. R_m , R_1 and R_3 are connected in a wye
- ☒ E. R_m , R_2 and R_x are connected in a wye

Delta-to-Wye Equivalent Circuits (1)

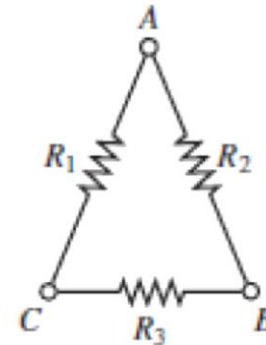
Delta-to-Wye (Pi-to-Tee) Equivalent Circuits

- A delta-connected configuration can be replaced by an equivalent wye-connected configuration, and vice versa.



$$R_{EQ} \equiv \frac{v_s}{i_s}$$

$$\Leftrightarrow$$



$$R_{AB} = R_x + R_y$$

Wye network

$$= R_2 \parallel (R_1 + R_3) = \frac{R_2(R_1 + R_3)}{R_1 + R_2 + R_3}$$

Delta network

$$R_{AC} = R_x + R_z$$

Wye network

$$= R_1 \parallel (R_2 + R_3) = \frac{R_1(R_2 + R_3)}{R_1 + R_2 + R_3}$$

Delta network

$$R_{BC} = R_y + R_z$$

Wye network

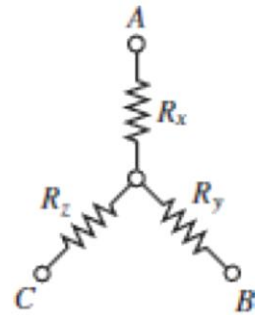
$$= R_3 \parallel (R_1 + R_2) = \frac{R_3(R_1 + R_2)}{R_1 + R_2 + R_3}$$

Delta network

Delta-to-Wye Equivalent Circuits (2)

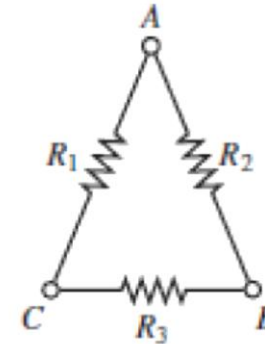
Delta-to-Wye (Pi-to-Tee) Equivalent Circuits

- A delta-connected configuration can be replaced by an equivalent wye-connected configuration, and vice versa.



$$R_{EQ} \equiv \frac{v_s}{i_s}$$

$$\Leftrightarrow$$



$$R_x = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$R_y = \frac{R_2 R_3}{R_1 + R_2 + R_3}$$

$$R_z = \frac{R_1 R_3}{R_1 + R_2 + R_3}$$

$$R_1 = \frac{R_x R_y + R_x R_z + R_y R_z}{R_y}$$

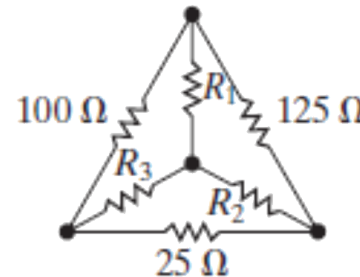
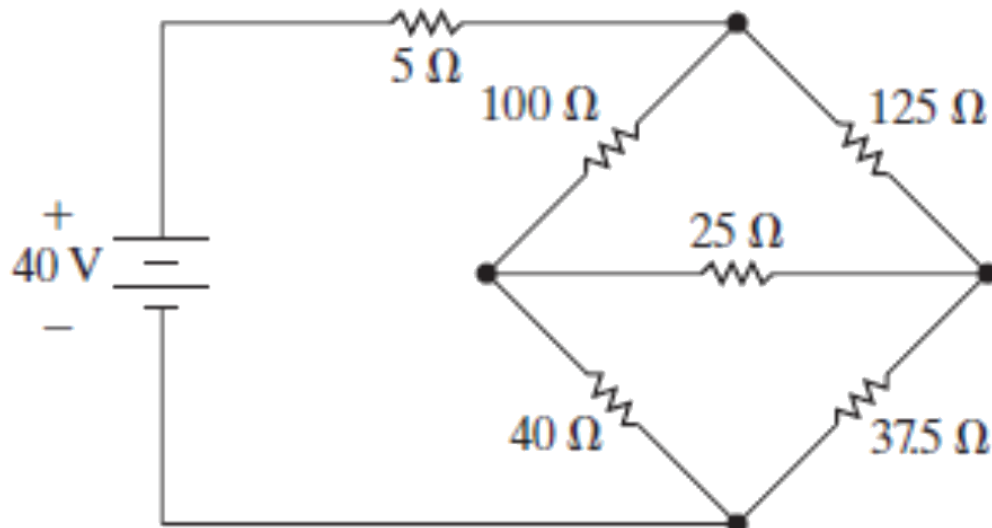
$$R_2 = \frac{R_x R_y + R_x R_z + R_y R_z}{R_z}$$

$$R_3 = \frac{R_x R_y + R_x R_z + R_y R_z}{R_x}$$

Example: Delta-to-Wye Conversion (1)

Delta-to-Wye Equivalent Circuits: Example

- For the circuit shown here, find the current in the voltage source.
- To do this, find the equivalent resistance seen by the voltage source. Start by replacing the upper delta with its equivalent wye.



$$R_1 = \frac{(100)(125)}{25 + 100 + 125} = 50 \Omega$$

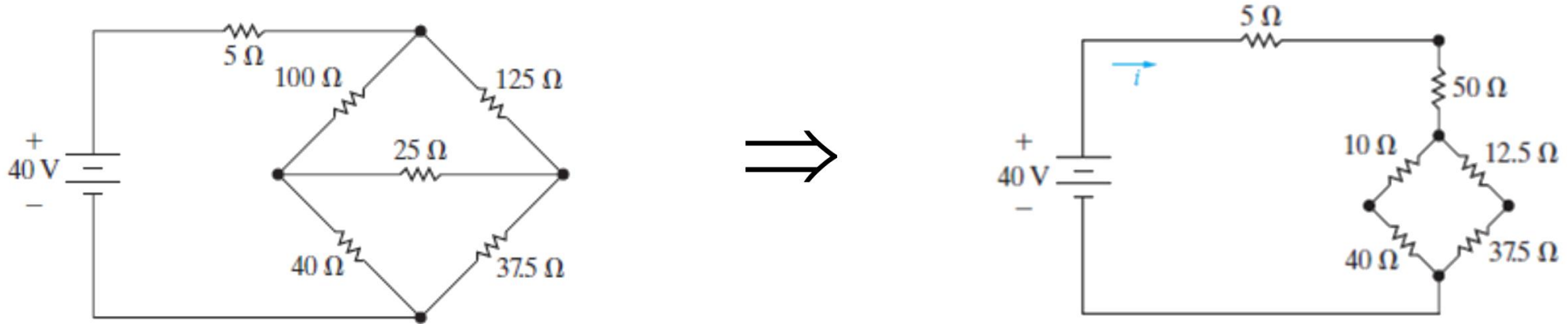
$$R_2 = \frac{(100)(25)}{25 + 100 + 125} = 12.5 \Omega$$

$$R_3 = \frac{(25)(100)}{25 + 100 + 125} = 10 \Omega$$

Example: Delta-to-Wye Conversion (2)

Delta-to-Wye Equivalent Circuits: Example (continued)

- For the circuit shown here, find the current in the voltage source.



The equivalent resistance seen by the source is

$$R_{eq} = 5 + 50 + [(10 + 40) \parallel (12.5 + 37.5)] = 5 + 50 + [50 \parallel 50] \\ = 5 + 50 + 25 = 80 \Omega$$

Therefore, the current in the voltage source is

$$i = \frac{40}{80} = 0.5 \text{ A.}$$

2.4 Linear Networks and the Principle of Superposition (1)

- In general, the criteria for a linear function are:
 - Superposition: If $y_1 = f(x_1)$ and $y_2 = f(x_2)$, then $y_1 + y_2 = f(x_1 + x_2)$.
 - Homogeneity: If $y = f(x)$, then $\alpha y = f(\alpha x)$.
- How to determine if a network is linear or not?

Any network composed of linear elements only is itself linear. Common linear elements are ideal sources, resistors, capacitors, and inductors.

→ then, the principle of superposition is valid

2.4 Linear Networks and the Principle of Superposition (2)

Principle of superposition:

- Each independent source contributes to each voltage and current in the circuit.
- Contributions of one source are independent of other sources.
- Each voltage and current in a circuit with N independent sources is the sum of N component voltages and N component currents, respectively.

2.4 Linear Networks and the Principle of Superposition (3)

Principle allows problem to be divided into simpler problems.

Turn off all independent sources except one.

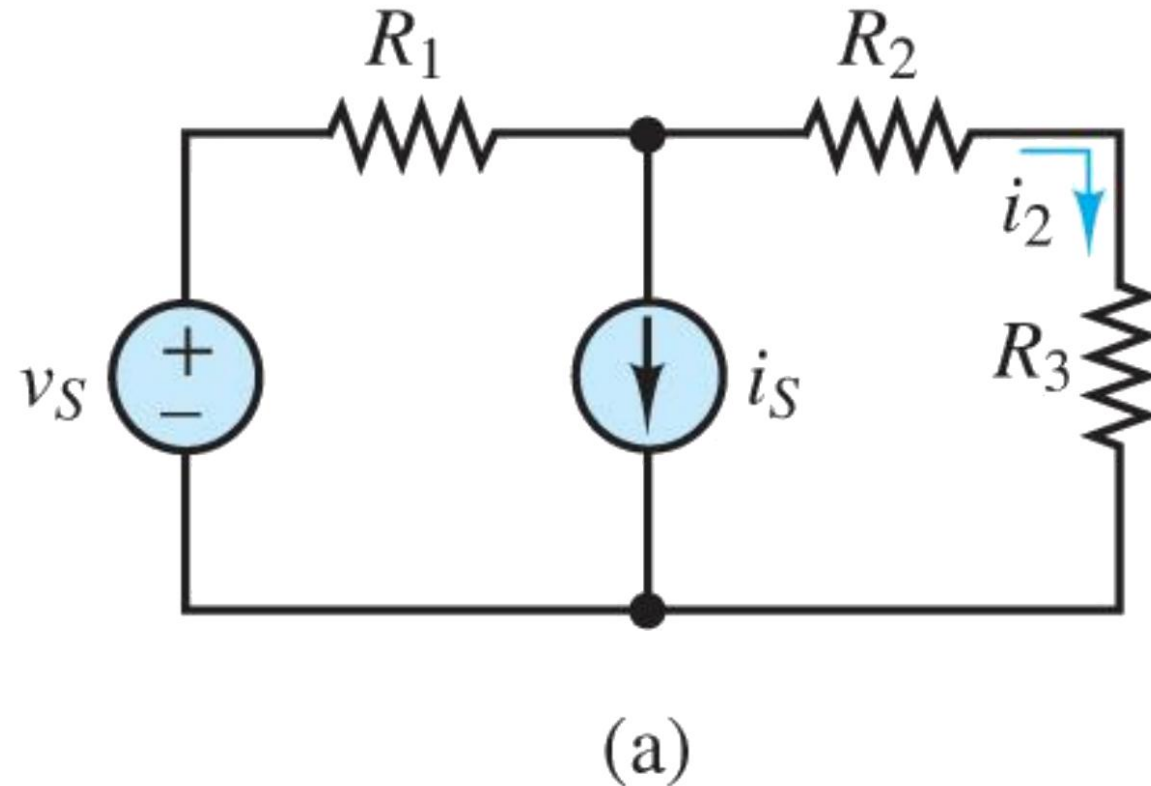
- Replace each zero source with a short- or open-circuit.

Solve for voltages and currents due to the remaining independent source.

Repeat process for each source.

Example 2.7: Principle of Superposition (1)

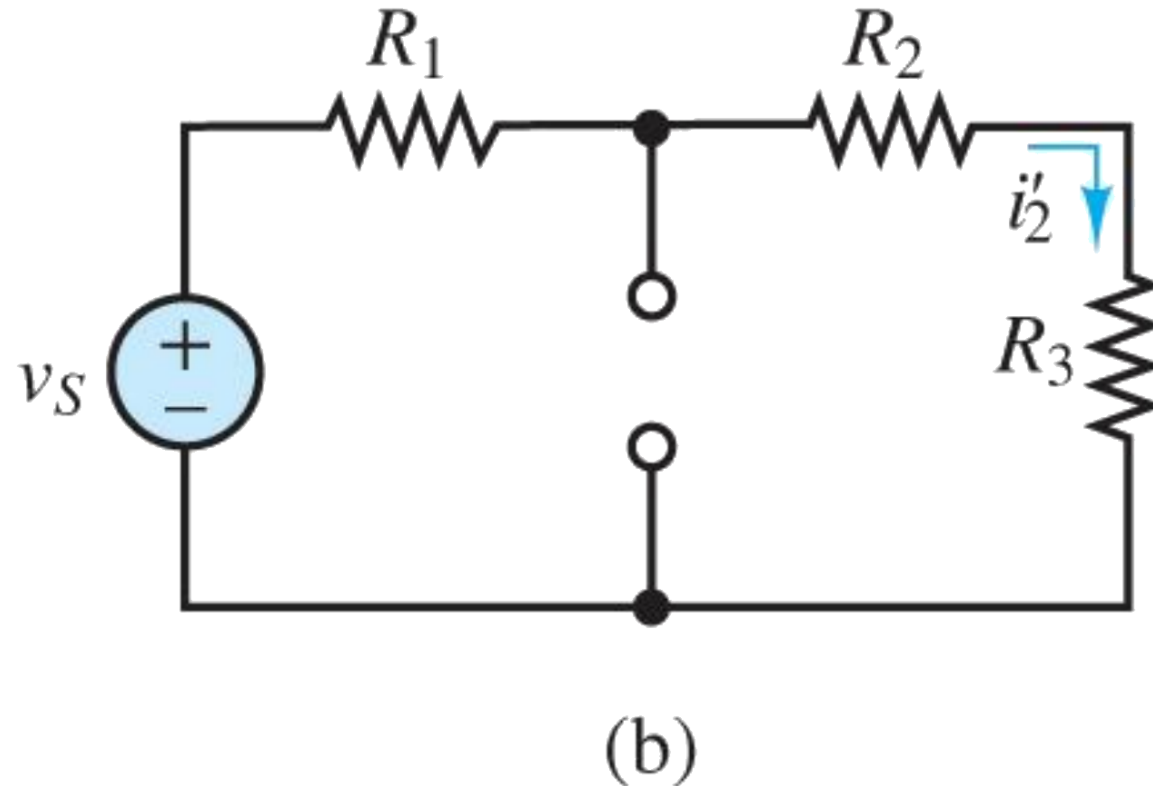
- Problem: determine current i_2 using the principle of superposition.
- Given: $v_S = 10\text{ V}$, $i_S = 2\text{ A}$, $R_1 = 5\ \Omega$, $R_2 = 2\ \Omega$, $R_3 = 4\ \Omega$.



Example 2.7: Principle of Superposition (2)

Turn off current source and replace it with an open circuit.

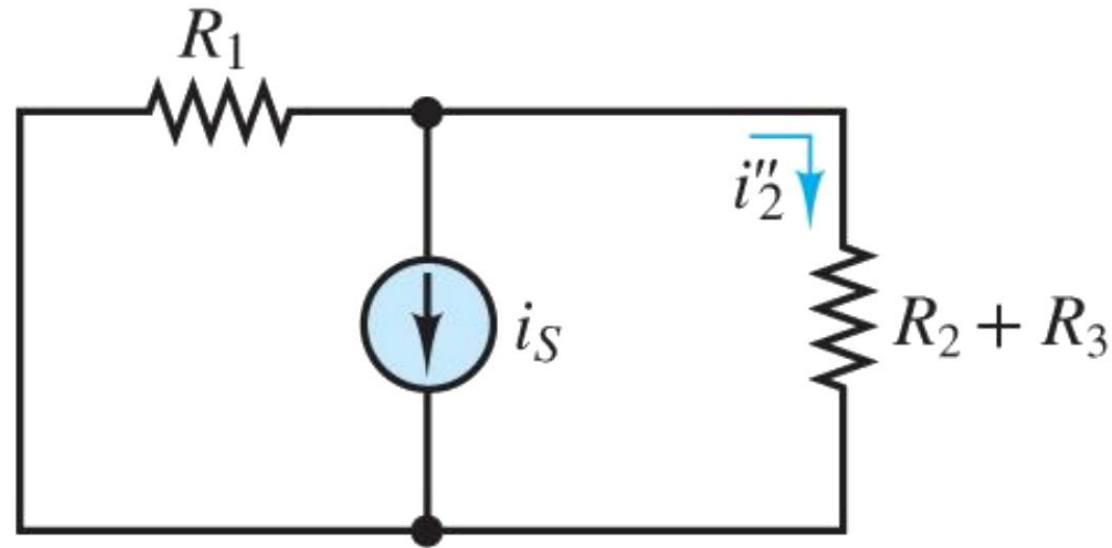
- Solve for total series resistance and use Ohm's law to find $i'_2 = 0.909$ A.



Example 2.7: Principle of Superposition (3)

Turn off the voltage source and replace it with a short circuit.

- Resulting circuit consists of three parallel branches.
- Use current division to find $i''_2 = -0.909 \text{ A}$.

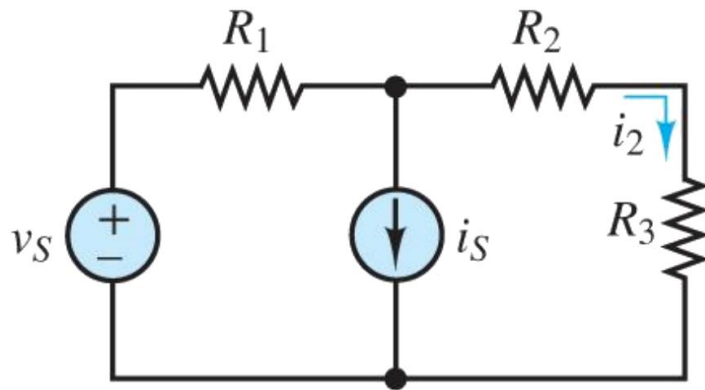


(c)

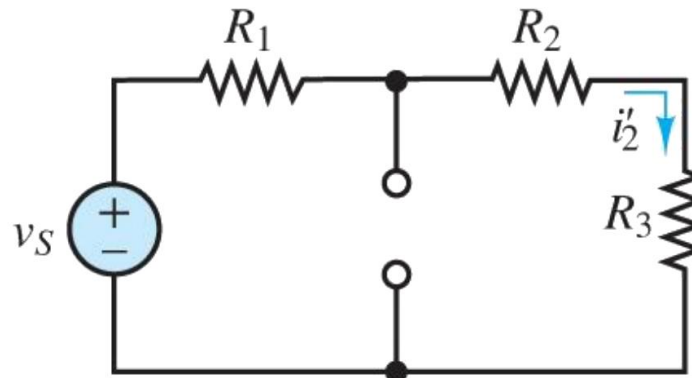
Example 2.7: Principle of Superposition (4)

- Sum the two components to find:

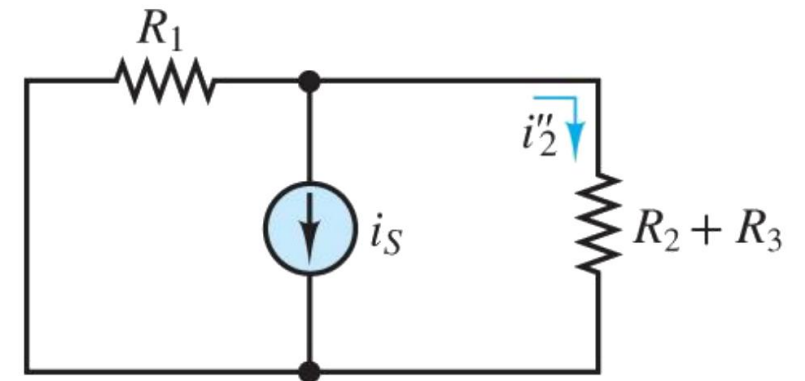
$$i_2 = i'_2 + i''_2 = 0A$$



(a)

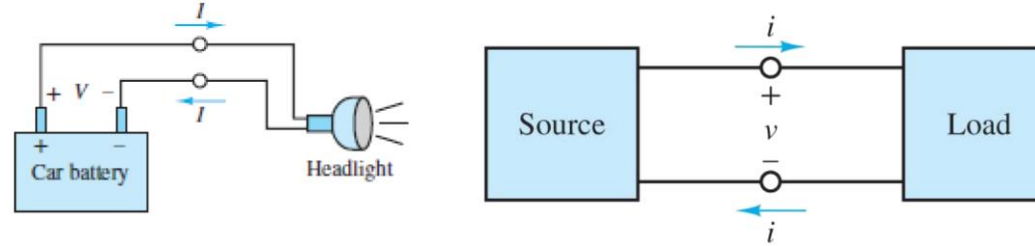


(b)



(c)

Thévenin and Norton Equivalent Networks (1)

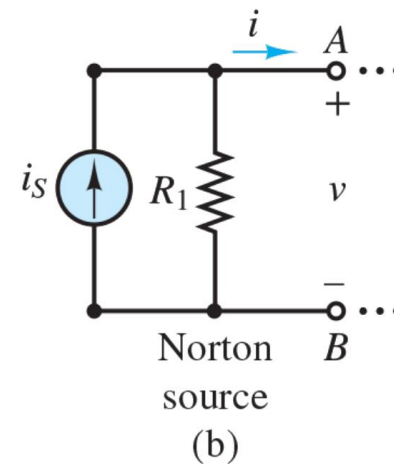
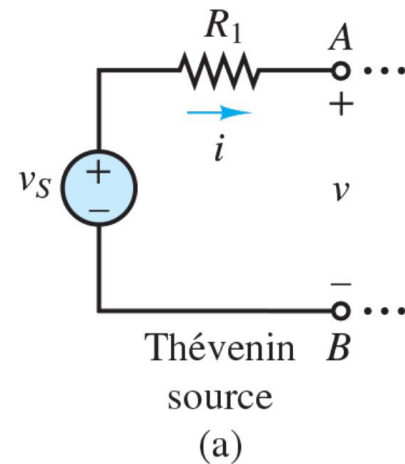
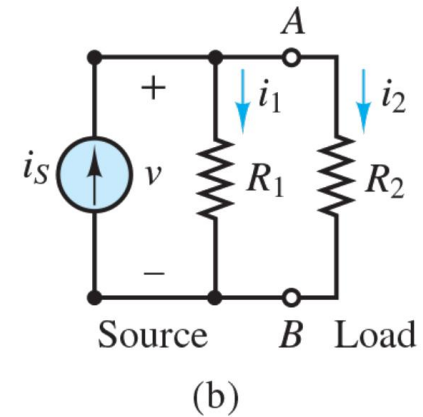
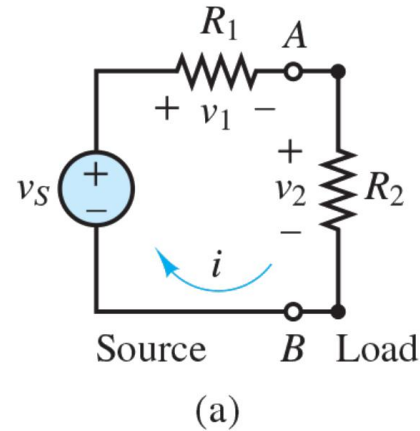


Divide a circuit into two parts: the source and the load.

- Load
 - The circuit element or segment of interest. It consumes the provided energy (can be represented as a simple resistor).
- Source
 - Source is everything not included in the load.
 - Concept different from ideal independent and dependent voltage and current sources.
 - The circuit provides energy (can be represented as a THÉVENIN or NORTON source).

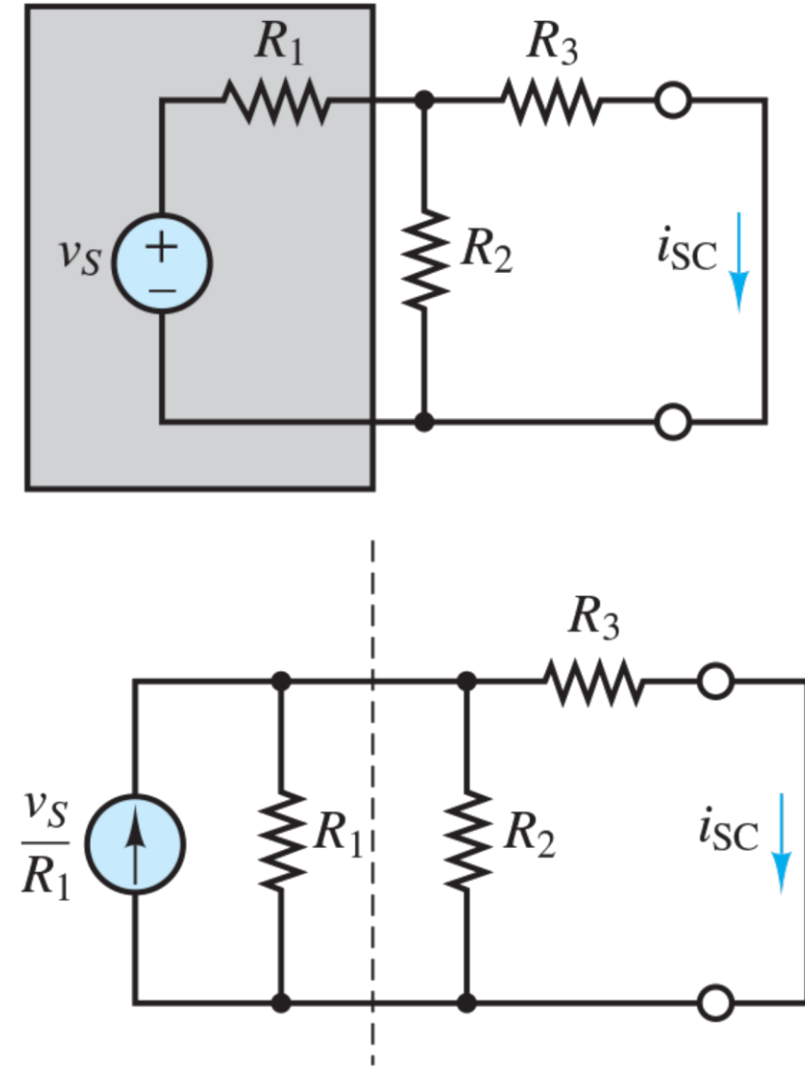
Thévenin and Norton Equivalent Networks (2)

- Perspective leads to graphical solutions.

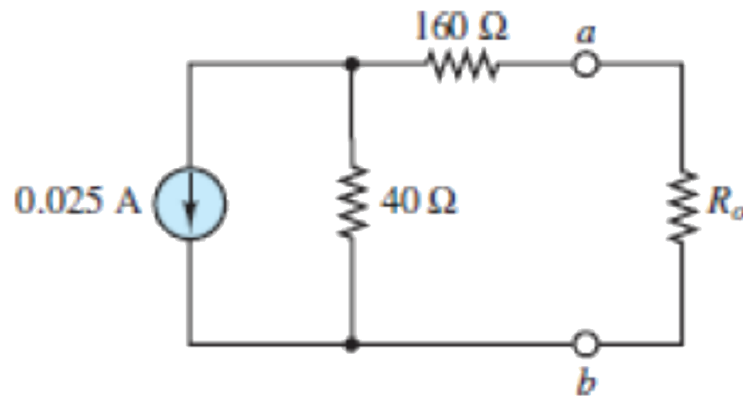
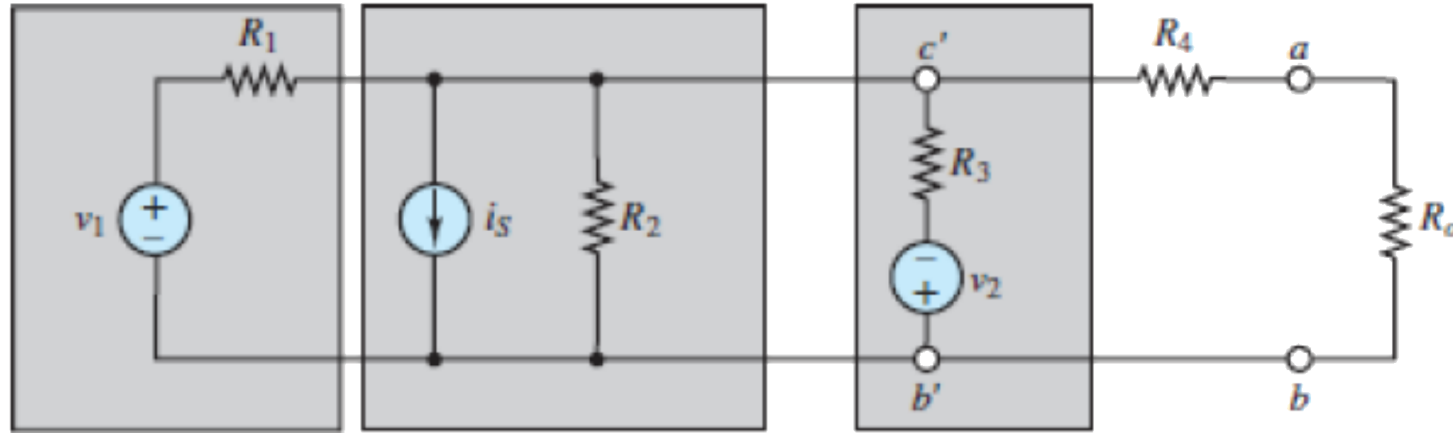


Thévenin and Norton Equivalent Networks (3)

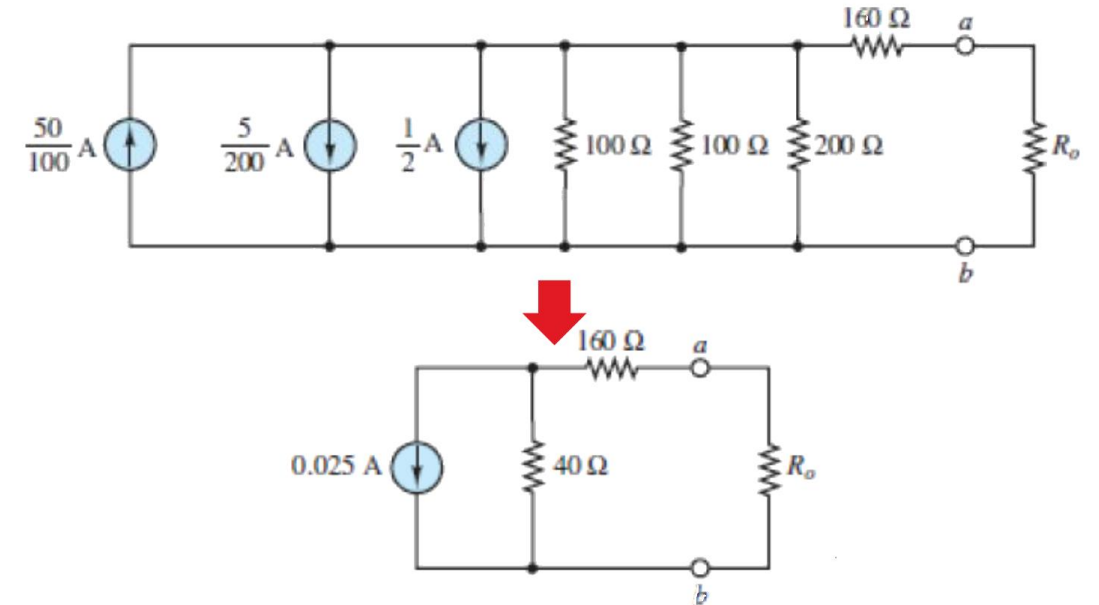
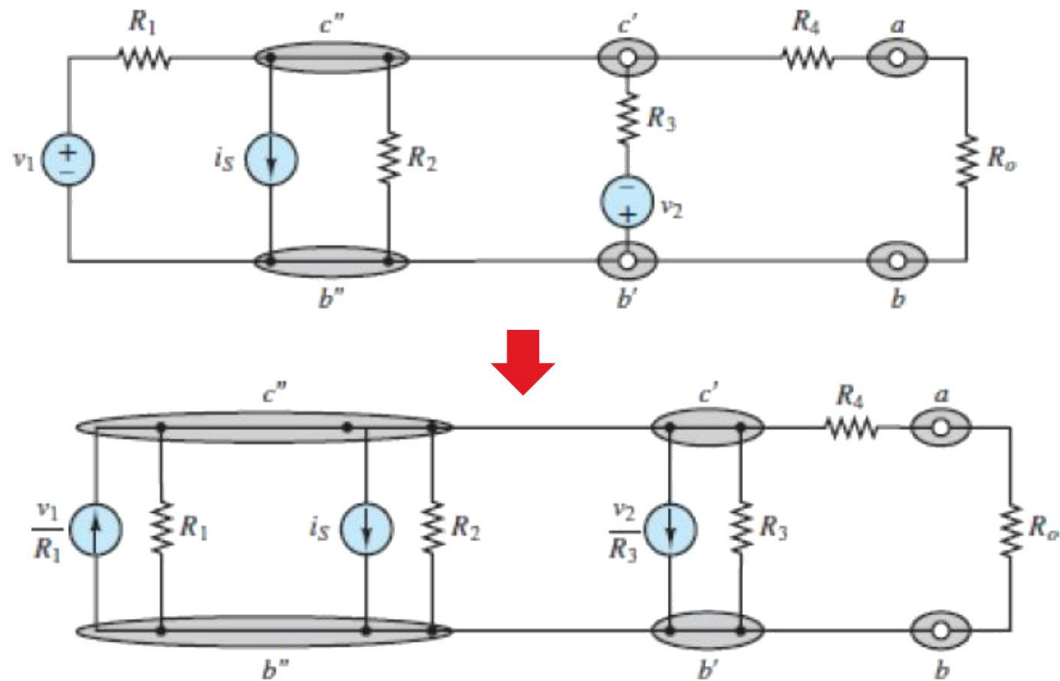
A Thévenin source can be transformed into a Norton source, and vice versa.



Thévenin and Norton Equivalent Networks (4)



Thévenin and Norton Equivalent Networks (5)



Homework

Problems

- 1.16, 1.17, 1.25, 1.30, 1.38
- 2.7, 2.11, 2.19, 2.27, 2.42